1. Introduction and Background

1.1 Overview

Introduction

The rapid advancement of digital technologies has fundamentally transformed recruitment and hiring processes across the world. Traditional manual screening of curricula vitae has increasingly been replaced by automated Applicant Tracking Systems, online job portals, and algorithm-driven recommendation platforms. As a result, job seekers are now required not only to possess strong technical competencies but also to present their qualifications in well-structured, machine-readable formats that align with industry expectations. A poorly formatted or weakly written CV can significantly reduce a candidate's chances of being shortlisted, regardless of actual skill level.

Within Sri Lanka, this transformation is particularly visible in the Information Technology sector, where competition for internships and entry-level software development positions continues to grow each year. Universities produce a large number of IT graduates, yet many struggle to convert academic achievements into professional CVs that reflect industry requirements. Common issues include weak organization, missing technical keywords, lack of project evidence, absence of GitHub portfolios, and inclusion of unnecessary information such as school-level examination results. These weaknesses are further amplified when CVs are processed by automated screening systems, which rely heavily on keyword matching and standardized layouts.

In parallel, the job-search process itself has become increasingly complex. Hundreds of vacancies are advertised daily on Sri Lankan platforms such as TopJobs.lk and Ikman.lk, each requiring different technical skills, experience levels, and role specializations. Manually reviewing these opportunities is time-consuming and often impractical for students and graduates, leading many to overlook positions that closely match their profiles. Consequently, there is a growing need for intelligent systems that can analyze candidate qualifications and automatically recommend suitable job opportunities based on real market demand.

Although several international CV analysis and job recommendation tools exist, most are designed for Western labor markets and often require expensive subscriptions. These systems rarely account for Sri Lankan recruitment practices, cultural expectations, or local job portals, limiting their usefulness for local users. This gap highlights the necessity for a localized, accessible, and integrated platform that simultaneously improves CV quality and supports informed job search decision making.

Background

In recent years, recruitment has changed because of new digital systems and online job platforms. Many companies now use Applicant Tracking Systems and automated tools to screen CVs before human review. These systems check keywords, skills, education, and formatting. This helps companies save time, but it creates problems for job seekers who do not understand how these systems work.

In Sri Lanka, the IT industry is growing, and many students apply for internships and entry level jobs every year. However, many students do not know how to prepare a good CV that meets industry requirements. Some CVs have poor structure, missing technical skills, no project details, or no GitHub links. Some students also include school exam results that are not needed for IT jobs. Because of these problems, good candidates can be rejected early in the hiring process.

At the same time, there are many IT job vacancies on Sri Lankan job websites such as TopJobs.lk and Ikman.lk. Each job asks for different skills and experience. Students must read many job posts to find the best one for them, which is not practical. Because of this, students may miss good opportunities that match their skills.

Although there are CV checking and job recommendation systems, most of them are made for foreign job markets. They are often expensive. This situation shows the need for a system that improves CV quality and recommends suitable jobs for the Sri Lankan IT job market. The proposed research develops a system that uses local employer requirements and real job data from Sri Lankan websites to help students apply for jobs more successfully.

1.2 Project Scope

This research focuses on developing and evaluating a system that improves CV quality and recommends suitable job opportunities for the Sri Lankan IT job market. The scope is defined to keep the objectives clear, focused, and useful.

The system will:

- Focus mainly on Information Technology (IT) jobs, including software development roles, internships.
- Work only with CVs written in English and uploaded as PDF.
- Analyze CV content, structure, formatting, technical skills, and professionalism.
- Provide feedback to help users improve their CVs.
- Recommend jobs by matching CV details with job requirements taken from Sri Lankan job websites such as TopJobs.lk.

The study will not include:

- CVs written in Sinhala or Tamil languages.

By limiting the scope in this way, the system can focus on the local IT job market and deliver accurate and practical results for Sri Lankan students and job seekers.

1.3 Motivation for the Study

The main motivation for this study comes from the problems faced by Sri Lankan IT students and job seekers when preparing CVs and searching for suitable jobs. Many students do not receive enough guidance on how to create a professional CV that meets industry requirements. Because of this, their CVs often have weak structure, missing technical skills, poor formatting, and limited project details or GitHub links.

Another important reason for this study is the large number of IT job vacancies available on Sri Lankan job websites such as TopJobs.lk and Ikman.lk. Each job asks for different skills and experience. Reading and comparing many job posts takes a lot of time and effort, which is not

practical for students. Because of this, students may miss good job opportunities that match their qualifications.

Many existing CV checking and job recommendation systems are expensive and require paid subscriptions. These systems are mostly made for foreign job markets and do not follow Sri Lankan recruitment practices or use local job websites. Therefore, Sri Lankan students cannot depend on these systems to improve their CVs or find suitable jobs.

This research is motivated by the need to create a free or low-cost system that improves CV quality and recommends suitable job opportunities for the Sri Lankan IT job market. The proposed system aims to help students prepare better CVs, understand job market requirements, and apply for jobs with more confidence.

1.4 Rationale and Justification

The reason why developing a CV enhancement and job opportunity recommendation system is based on both technical necessity and the practical requirements of the Sri Lankan IT software engineering job market. While automation in recruitment is increasing globally, existing solutions fail to address the specific gap between academic output and industry expectations in Sri Lanka.

The proposed system is justified by its use of advanced Natural Language Processing (NLP) and Optical Character Recognition (OCR) technologies, specifically the EasyOCR engine and PyMuPDF. Unlike standard keyword-matching tools that often fail to read complex PDF layouts or images, the integration of EasyOCR ensures that even non-selectable text is accurately captured. Furthermore, the use of TF-IDF (Term Frequency-Inverse Document Frequency) and Cosine Similarity provides a mathematically robust way to measure the alignment between a candidate's profile and a job description. This technical approach is superior to simple boolean searches as it understands the weight and importance of specific technical skills within a document.

A significant justification for this research is the current lack of support for the Sri Lankan job ecosystem. Most commercial career tools are optimized for international platforms like LinkedIn

and do not index local portals such as TopJobs.lk or Ikman.lk. This system is specifically designed to bridge this gap by:

- Local Data Integration: Scraping and analyzing vacancies directly from Sri Lankan job sites to ensure recommendations are relevant to local graduates.
- Industry-Specific Alignment: Mapping technical keywords that are currently trending within the ICT sector specializing in the software engineering field, rather than relying on generic global trends.

The utilitarian benefits of this system are substantial for both graduate and undergraduate students. For students, the system reduces the time spent on manual job searching and CV analyzing from several hours to a few minutes of automated analysis. By providing a "CV Score" and specific suggestions for improvement, it empowers students to fix errors before they apply, significantly increasing their chances of passing initial screenings.

From an institutional perspective, the system utilizes reliable, open-source Python libraries and frameworks like Flask, which have been rigorously tested in academic and commercial environments. This ensures that the platform is not only high-performing but also cost-effective and scalable for local universities to implement for their students. Ultimately, this research provides a practical solution to the high unemployment or underemployment rates among ICT graduates by aligning their presentation with the real-world demands of Sri Lankan employers.

1.5 Research Objectives

This research aims to develop and evaluate a system that improves CV quality and recommends suitable job opportunities for the Sri Lankan IT job market. The objectives are defined to make sure the study remains focused and produces practical and useful outcomes for students, graduates, and employers.

1.5.1 Main Objective

The main objective of this study is to develop a CV Enhancement and Job Opportunity Recommendation System that follows Sri Lankan IT industry requirements and helps users find the best possible job opportunities from the local job market.

1.5.2 Specific Objectives

- **Create a CV analysis system**

  The system will read CVs uploaded as PDF or Word files and automatically find important information like skills, work experience, education, and qualifications. After studying the CV, it will create personalized advice about the mistakes and errors that were made. This solves a big problem because right now, there are no automatic CV checking tools made specifically for Sri Lankan ICT Software Development job seekers.

- **Develop a job recommendation system**

  The system will take skills, work experience, and more data as input. Then compare them with IT job requirements from popular Sri Lankan job websites like TopJobs.lk and Ikman.lk. The system will then suggest jobs that are perfect for the qualifications, doing much more than just looking for matching words between CV and job descriptions.

- **Establish a user-centric interface and feedback system**

  Develop a simple, accessible interface where users, especially university students and IT job seekers, can upload CVs, receive automatic feedback, and view job recommendations. The interface will ensure actionable career-development guidance and ease of use for diverse Sri Lankan users, addressing barriers created by complex or premium international platforms.

1.6 The Conceptual, Theoretical, and Practical Significance of Research

This research is important at three levels: conceptual, theoretical, and practical.

Conceptual Significance

From a conceptual point of view, this research introduces a combined approach where CV enhancement and job recommendation work together in a single system. Instead of treating these two tasks separately, the system first improves CV quality and then uses the improved CV data to recommend suitable jobs. This helps create a clearer and more effective recruitment-support process for the Sri Lankan IT job market.

Theoretical Significance

From a theoretical point of view, this research contributes to knowledge in the areas of recruitment systems and information systems by showing how automated CV analysis, rule-based scoring, and job matching can be adapted for a local job market. The study also explains how employer requirements collected through HR surveys and job listing analysis can be used to define system rules and evaluation criteria. These ideas can support future research that focuses on localized employment-support systems in developing countries.

Practical Significance

From a practical point of view, the system provides direct benefits to Sri Lankan IT students and job seekers. It helps users identify weaknesses in their CVs, understand current job market skill demands, and find suitable job opportunities without depending on expensive international tools. The system also helps employers by improving the overall quality of submitted CVs, which can reduce recruitment time and effort. Overall, the proposed system supports more efficient hiring and better career guidance within the Sri Lankan IT industry.

1.7 Expected Outcomes

The expected outcomes of this research are to produce a working system that improves CV quality and recommends suitable job opportunities for the Sri Lankan IT job market.

Specifically, this study aims to achieve the following outcomes:

- Develop a fully functional CV Enhancement and Job Opportunity Recommendation System tailored to Sri Lankan IT recruitment practices.
- Provide clear and useful feedback to users so they can improve CV structure, formatting, technical skills, and professionalism.
- Generate accurate job recommendations based on real job data collected from Sri Lankan job websites such as TopJobs.lk and Ikman.lk.
- Identify skill gaps by comparing user profiles with current job market demands and guide users toward relevant learning areas.

- Reduce the time and effort required for students and graduates to search for suitable job opportunities.
- Increase user confidence and employability by helping them submit stronger and more industry-ready CVs.

These outcomes support the overall goal of improving recruitment success for Sri Lankan IT job seekers and providing a practical decision-support tool for career development.

2. Literature Review

2.1 Critical Evaluation

Historically, curriculum vitae (CVs) were simple documents presenting personal details, qualifications, and work experience in a chronological format (Jain et al., 2019). With the evolution of digital recruitment platforms, the role of CVs has significantly transformed. Modern hiring processes increasingly depend on automated tools such as Applicant Tracking Systems (ATS), which filter resumes based on predefined criteria and keywords (Liu et al., 2020).

Early CV optimization techniques primarily focused on formatting and presentation, including layout design and visual appeal (Kumar et al., 2018). However, these approaches lacked the ability to evaluate content quality effectively. The integration of Natural Language Processing (NLP), Machine Learning (ML), and Artificial Intelligence (AI) has enabled more advanced systems capable of parsing resumes, extracting meaningful information, and matching candidates with job opportunities based on both syntactic and semantic analysis (Ye et al., 2020).

This section critically reviews existing research in CV enhancement, resume parsing, and job recommendation systems, highlighting their strengths, limitations, and research gaps.

CV Enhancement Systems

CV enhancement systems have evolved from simple rule-based approaches to more advanced machine learning-driven solutions. Traditional methods often rely on classification algorithms such as Naive Bayes and K-Nearest Neighbours (KNN) to identify patterns in resumes and provide recommendations (Aggarwal and Aggarwal, 2021).

For instance, Onukwugha et al. (2024) developed a resume optimization model trained on a dataset of over 10,000 resumes. The system applies preprocessing techniques such as removing noise (URLs, punctuation, and whitespace) and uses KNN to identify similar CV patterns, achieving an accuracy of 87%. While this approach demonstrates reasonable performance, it suffers from limited generalization capability. The system depends heavily on existing patterns, making it ineffective when encountering new or unique CV formats. Furthermore, it lacks detailed, step-by-step feedback and does not integrate job recommendation features.

Similarly, intelligent resume screening systems have been developed using neural network models combined with NLP techniques. Abhishek et al. (2025) introduced a system based on a Multi-Layer Perceptron Neural Network (MLPNN) with NLP libraries such as NLTK and Pyresparser. This system achieved an accuracy of 85% and provided personalized feedback and analytics for users. However, despite these improvements, the system does not connect users with real job opportunities and lacks localization for specific job markets.

Another approach by Daryani et al. (2020) utilized NLP techniques such as tokenization and Named Entity Recognition (NER), combined with TF-IDF and cosine similarity for resume-job matching. While effective for basic matching, this system relies heavily on keyword-based techniques and was tested on a very limited dataset, reducing its reliability and real-world applicability. Additionally, it does not provide feedback to improve CV quality.

Overall, existing CV enhancement systems face common limitations, including reliance on keyword matching, lack of semantic understanding, absence of detailed feedback mechanisms, and failure to integrate job recommendation features. These gaps highlight the need for more intelligent and adaptive systems capable of understanding context and providing actionable guidance.

To address these limitations, recent research trends emphasize the use of transformer-based models and Large Language Models (LLMs), which enable semantic analysis and flexible evaluation of CVs. Unlike traditional systems, these models can understand meaning rather than relying solely on pattern matching, making them more suitable for handling diverse CV formats.

Resume Parsing and NLP Techniques

Resume parsing plays a crucial role in extracting structured information from unstructured CV documents. Early approaches relied on basic NLP techniques such as tokenization, stemming, and keyword extraction (Pandey et al., 2021). However, these methods often struggled with variations in language and formatting.

Recent advancements have introduced deep learning-based models, particularly transformer architectures such as BERT, which significantly improve contextual understanding (Devlin et al., 2018). Hybrid approaches combining tools like SpaCy with transformer models have

demonstrated high accuracy in extracting key information such as skills, education, and experience (Bhoir et al., 2023).

Machine learning algorithms such as Support Vector Machines (SVM), Long Short-Term Memory (LSTM), and Random Forest have also been widely used for classification and information extraction tasks (Liu et al., 2020). These approaches improve automation and efficiency in recruitment processes.

However, despite these advancements, challenges remain. Many systems still rely on keyword-based representations, which fail to capture semantic nuances such as synonyms and contextual meaning. This limitation can result in the rejection of qualified candidates due to differences in wording (Jirjees et al., 2025). Additionally, deep learning models often require large datasets and high computational resources, limiting their accessibility.

Job Recommendation Systems

Job recommendation systems play a crucial role in modern recruitment by automatically matching candidates with suitable job opportunities based on their profiles, skills, and preferences. Traditional approaches primarily rely on content-based filtering and collaborative filtering techniques. Content-based methods compare candidate CVs with job descriptions using techniques such as Term Frequency–Inverse Document Frequency (TF-IDF) and cosine similarity, while collaborative filtering recommends jobs based on the behavior and preferences of similar users (Guo et al., 2016).

Hybrid recommendation models, which combine both approaches, have been widely adopted to improve performance. For instance, Kumar et al. (2022) developed a technical job recommendation system that integrates content-based filtering and collaborative filtering. Their system collects job data using multiple sources, including job portals, company websites, and APIs, enabling real-time job recommendations. While the content-based approach alone achieved relatively low performance (37.74%), collaborative filtering improved the results (59.78%). The hybrid model significantly enhanced the overall recommendation performance.

Despite these strengths, such systems exhibit notable limitations. They are typically restricted to specific datasets in this case, jobs from only 100 global companies and do not assess the quality

of a candidate's CV before generating recommendations. As a result, users with poorly structured or incomplete CVs may receive irrelevant job suggestions without guidance on improving their applications.

Recent advancements in machine learning and deep learning have further improved job recommendation systems. Transformer-based models such as Bidirectional Encoder Representations from Transformers (BERT) enable semantic understanding of both resumes and job descriptions. Jirjees et al. (2025) demonstrated that BERT-based models, along with Support Vector Machines (SVM), can achieve high accuracy (up to 94%) in job-candidate matching tasks. These models outperform traditional keyword-based approaches by capturing contextual meaning rather than relying solely on exact keyword matches.

However, these systems also present practical challenges. Many models focus primarily on skill-based matching, ignoring other critical factors such as work experience, education level, and career progression. This limitation makes them less effective for fresh graduates and internship seekers who may lack extensive skill sets. Additionally, transformer-based models require substantial computational resources, making them expensive and difficult to deploy in real-world environments with limited infrastructure. Furthermore, many studies rely on static or pre-collected datasets rather than real-time job data, raising concerns about their applicability in dynamic job markets.

More recent research has explored the integration of multiple AI models to enhance recommendation quality. Ari (2024) proposed a system combining BERT and Generative Pre-trained Transformer (GPT) models, introducing a "Job Desirability Score" that incorporates factors such as company ratings, job type, and user preferences. This approach provides a more comprehensive evaluation of job opportunities compared to traditional methods. However, the system is limited by its short data collection period and domain specificity, as it focuses only on software engineering roles, reducing its generalizability across industries.

Similarly, Grace et al. (2025) developed a machine learning-based job recommendation system using NLP techniques, TF-IDF vectorization, and text preprocessing to generate real-time personalized job suggestions. While effective in delivering recommendations, the system assumes that candidates already possess well-structured and professionally written CVs. This

assumption represents a significant limitation, particularly in contexts such as Sri Lanka, where many graduates lack proper guidance in CV preparation and often include irrelevant or incomplete information.

A common limitation across most job recommendation systems is the absence of integration with CV enhancement mechanisms. These systems generally assume that candidate profiles are already optimized, failing to address issues related to CV quality, structure, and content. Consequently, even highly advanced recommendation algorithms may produce suboptimal results if the input CV is poorly constructed.

To overcome these limitations, this research proposes an integrated approach that combines CV enhancement with job recommendation. By incorporating semantic analysis through Large Language Models (LLMs) and continuously collecting real-time job data from Sri Lankan job portals, the system ensures both accurate CV evaluation and relevant job matching. Additionally, the inclusion of localized knowledge derived from HR surveys enables the system to provide recommendations that are aligned with the expectations of the Sri Lankan IT industry. This holistic approach addresses the gaps identified in existing systems and enhances the overall effectiveness of the job-seeking process.

Analysis of existing applications for CV Enhancement and Job Opportunity Recommendations

Beyond academic research, several commercial platforms are widely used for CV enhancement and job recommendation. Popular tools include JobScan, Resume Worded, Enhance CV, Seveum, and LoopCV. These platforms provide useful features such as keyword optimization, formatting suggestions, and automated job matching. However, when these tools are analyzed in detail, several important limitations can be identified, especially in the context of the Sri Lankan job market.

JobScan is mainly designed to improve CVs by matching keywords with job descriptions. It helps users optimize their resumes to pass Applicant Tracking Systems (ATS), which are commonly used by large international companies. However, its recommendations are mostly based on global job standards and do not consider local recruitment practices. As a result, the

suggestions provided may not align with the expectations of Sri Lankan employers, reducing its practical usefulness for local job seekers.

Resume Worded focuses on improving CV quality by checking grammar, formatting, and content clarity. It provides a scoring system and general feedback to guide users. But requires an expensive premium subscription to access its full features. This makes it less accessible for students and fresh graduates. In addition, the feedback is often generic and does not provide detailed, step-by-step guidance tailored to specific industries or roles.

Enhance CV and Seveum primarily focus on improving the visual design and structure of resumes. They offer templates and formatting suggestions that make CVs look more professional. However, these platforms lack intelligent content analysis. They do not deeply evaluate the relevance of skills, projects, or experience based on job market demand. Therefore, while the CV may look visually appealing, it may still lack the content quality required to attract employers.

LoopCV is a job recommendation tool. It suggests opportunities based on user profiles. However, most of the job listings come from international platforms such as LinkedIn, Indeed, and Glassdoor. This creates a major limitation for users, as many local companies mostly does not advertise their vacancies on these platforms. Instead, they rely on local job portals such as TopJobs.lk and Ikman.lk. As a result, users may miss relevant local job opportunities when relying on such systems.

These tools are mostly designed for a global audience and less consider country-specific factors such as local industry demands. Furthermore, they do not validate practical elements in CVs, such as checking whether GitHub project links are working, evaluating the professionalism of CV content, or identifying skill gaps based on recommending jobs.

3. Research Methodology

This chapter explains the methodology used to design, develop, and evaluate the proposed AI-assisted CV enhancement and validation system, which is specially designed for the Sri Lankan IT job market. The purpose of this system is to help students improve their CV quality and increase their chances of getting internships and entry-level IT jobs.

In this research, a mixed-methods approach was used. This means both quantitative and qualitative methods were applied. The quantitative part focuses on measuring the performance of the system, such as how accurately it detects CV components, how consistently it generates scores, and how reliable the evaluation results are. The qualitative part focuses on understanding the user experience, including how useful the feedback is, how easy the system is to use, and whether the recommendations are helpful for improving CV quality.

The research methodology is divided into six main phases. The first phase is data collection, where information was gathered from students and recruiters to understand real CV evaluation practices in Sri Lanka. The second phase is data preprocessing, where uploaded CV files are processed and converted into structured data that can be analyzed by the system. The third phase is system development and implementation, where the validation rules, AI integration, scoring system, and feedback generation modules were designed and implemented. The fourth phase is evaluation and scoring, where the system analyzes the CV using rule-based validation and AI-assisted validation to generate scores and feedback. The fifth phase is the job recommendation system development, where machine learning models and similarity-based techniques are used to match user CVs with relevant job opportunities and predict suitable job roles. The final phase is user validation, where the system results are tested and evaluated to ensure accuracy, fairness, and usefulness for students and job applicants.

3.1 Research Design

This study uses a mixed-methods research design, because the proposed CV validation system needs to be evaluated from both a technical and user perspective. The system must not only work correctly from a technical point of view, but it must also provide useful and meaningful feedback for students and job applicants.

The first part of the evaluation focuses on the objective performance of the system. This includes checking whether the platform can correctly detect important CV components such as contact information, skills, GitHub links, and professional summary. It also evaluates whether the system produces consistent scores for the same CV and whether the generated results are accurate, structured, and reliable.

The second part of the evaluation focuses on the human-centered quality of the platform. This means understanding how users, especially students and recruiters, experience the system. The study evaluates whether the feedback generated by the system is easy to understand, helpful for improving CV quality, fair to all users, and aligned with real expectations of Sri Lankan IT recruiters.

The main design principle of this research is to ensure that the CV evaluation system reflects real Sri Lankan recruitment practices rather than using generic international CV standards. To achieve this, data was collected from real stakeholders, including undergraduate students and IT recruiters. Their responses were analyzed to understand what recruiters expect in a CV and what common mistakes students make.

These findings were then converted into system validation rules, scoring criteria, and feedback generation logic. For the job recommendation component, the technical evaluation focuses on the accuracy and relevance of job matching. This includes evaluating how effectively the system matches user skills with job descriptions using similarity-based techniques, and how accurately the machine learning models predict suitable job roles based on the user's CV content. Additionally, the consistency of recommendations and the quality of skill-based matching are assessed. This ensures that the system provides practical and relevant recommendations that match the actual requirements of the Sri Lankan IT job market.

By integrating CV enhancement with job recommendation, the system provides a complete end-to-end solution, guiding users from CV improvement and recommending suitable job opportunities. This ensures that the system delivers practical, relevant, and actionable support tailored to the IT job market.

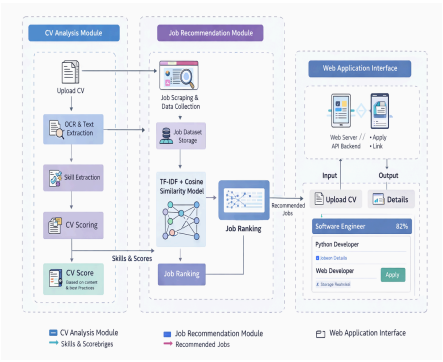
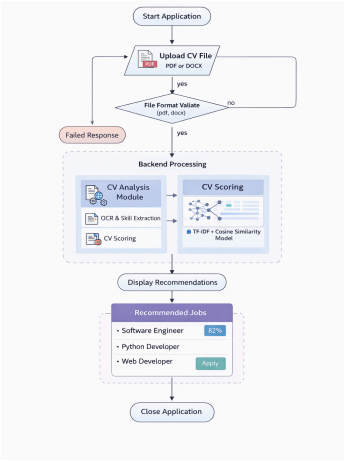
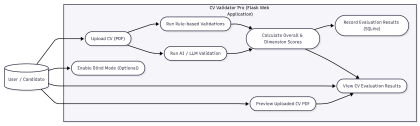
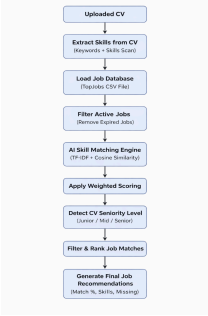


Figure 1: Overall Project Design







3.2 Data Collection

The data collection phase was conducted to ensure that the proposed CV validation and enhancement system reflects real-world recruitment practices in the Sri Lankan IT industry. Instead of relying on general international assumptions, this research collected data directly from Sri Lankan undergraduate students and IT recruiters. This approach helps ensure that the system's validation rules, scoring logic, and feedback are relevant, practical, and aligned with local hiring expectations.

Data was collected using two main methods: structured surveys and live observation of the Sri Lankan IT job market. The structured surveys were designed to capture both the student perspective and the recruiter perspective. In addition, job portals and recruitment platforms were observed to understand how internships and entry-level IT jobs are advertised and evaluated.

This combination of data sources helped ensure that the system design is based on real user needs and actual recruiter expectations.

3.2.1 Objectives of Data Collection

The main objective of the data collection phase was to understand how CV evaluation and job searching work in the Sri Lankan IT recruitment environment.

Specifically, this phase focused on identifying:

- how Sri Lankan undergraduate students search for internships and entry-level IT jobs,
- which job platforms and portals are most commonly used,
- common difficulties faced by students during job searching,
- common mistakes students make when preparing their CVs,
- how recruiters evaluate CVs during the screening process,
- what information recruiters consider most important,
- what makes a CV appear professional and competitive,
- and which factors cause CVs to be rejected.

These insights were essential for designing a CV validation system that provides meaningful feedback and accurate scoring.

Survey 01 – Student Internship Discovery Survey

The first survey was conducted among undergraduate students and recent graduates who had applied for internships or entry-level IT jobs. The purpose of this survey was to understand student job searching behavior and identify challenges related to CV preparation.

The survey collected information about how students search for internships, which platforms they use, and what difficulties they face during the application process.

The main objectives of this survey included:

- identifying commonly used job platforms such as LinkedIn, TopJobs.lk, Ikman.lk, company career websites, and personal referrals,
- understanding student job searching behavior and how they select job opportunities,
- identifying difficulties faced by students, such as lack of experience, lack of guidance, unclear job requirements, and difficulty finding internships,
- and evaluating student awareness of CV quality, structure, and professional standards.

The results of this survey provided important insights into student behavior and CV preparation challenges.

These findings contributed to the system design in several ways:

- helped identify which job portals should be supported and referenced by the system,
- highlighted common CV weaknesses such as missing GitHub links, poor formatting, weak skills sections, and lack of measurable achievements,
- and helped design targeted feedback messages to guide students in improving their CV quality.

This ensures that the system addresses real problems faced by Sri Lankan students.

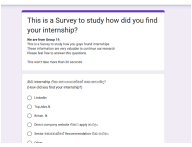
Survey 02 – Recruiter CV Evaluation Survey

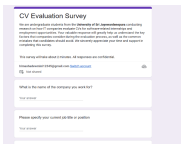
The second survey was conducted among HR professionals and technical recruiters working in leading Sri Lankan IT companies, including Dialog Axiata, Virtusa, IFS, WSO2, and CodeGen International.

The purpose of this survey was to understand how recruiters evaluate CVs and what they expect from IT job applicants.

The main objectives of this survey included:

- identifying what recruiters check first when reviewing a CV,
- understanding preferred CV structure and organization,
- identifying common reasons why CVs are rejected,
- evaluating the importance of technical skills, projects, internships, and certifications,
- understanding the importance of GitHub profiles and project repositories,
- identifying preferred CV length and formatting standards,





These insights were converted into specific validation checks implemented in the system, including:

- validating CV length to ensure it remains within the recommended range of one to two pages,
- checking for required sections such as education, skills, projects, and contact information,
- verifying the presence and accessibility of GitHub profile and repository links,
- detecting measurable achievements and quantifiable impact statements,
- validating professional email address format,
- checking formatting consistency and layout quality,
- and reducing the importance of school-level examination results such as O/L and A/L.

These validation rules were implemented using automated validation functions such as:

- check_cv_page_count
- check_contact_information
- validate_technical_keywords
- validate_github_links
- check_action_verbs
- check_quantifiable_achievements

By converting survey insights into system logic, the proposed platform ensures that CV evaluation reflects actual recruiter decision-making practices.

3.2.3 CV Upload Input Standard and Data Handling

The proposed system accepts curriculum vitae (CV) documents only in Portable Document Format (PDF). This decision was made to ensure consistency in document structure and layout. Unlike editable formats such as Microsoft Word, PDF files preserve formatting, alignment, fonts, and spacing across different devices and operating systems. Since PDF is the most commonly used format for professional CV submissions in the Sri Lankan IT recruitment process, restricting uploads to PDF helps maintain compatibility and reliable analysis.

To ensure secure and reliable file handling, several validation and safety mechanisms were implemented during the CV upload process.

First, the system performs file signature validation to confirm that the uploaded file is a genuine PDF document. Instead of relying only on the file extension (for example, ".pdf"), the system checks the binary header of the file to verify that it begins with the standard PDF signature %PDF. This prevents users from uploading malicious or unsupported files that are incorrectly renamed as PDF files. This approach improves system security and reduces the risk of harmful file execution.

Second, the system limits the maximum upload size to 16 megabytes. This size limit is sufficient to support professional CVs that may include images, graphics, or icons, while also protecting server resources. Restricting file size helps prevent excessive storage usage, performance degradation, and potential denial-of-service attacks caused by very large file uploads.

Third, uploaded CV files are stored only temporarily for analysis purposes. After the validation process is completed, the original uploaded file is automatically deleted from the server. This design ensures that sensitive personal information contained in CVs, such as phone numbers, email addresses, and academic details, is not stored permanently. By avoiding long-term storage of CV documents, the system follows basic data privacy principles and reduces the risk of data leakage.

To improve user experience, the system creates a temporary preview copy of the uploaded CV. This preview allows users to visually confirm that the correct document is being analyzed. Preview files are stored in a separate directory and are automatically deleted after a predefined time period. A time-based cleanup mechanism regularly removes outdated preview folders to prevent unnecessary storage accumulation.

3.3 Data Preprocessing

Curriculum vitae (CV) documents are considered unstructured data because they do not follow a fixed format. Different users may use different layouts, fonts, headings, and writing styles.

Because of this variation, the system cannot directly evaluate the CV without first converting it into a structured and standardized format.

Therefore, a preprocessing stage is required to extract, organize, and normalize CV content so that validation rules and AI-based analysis can be applied accurately and consistently. The preprocessing phase includes text extraction, structure detection, link verification, and markdown conversion.

3.3.1 Text and Structure Extraction

Once a CV is uploaded, the system uses the PyMuPDF library to open and read the PDF document. PyMuPDF is a reliable PDF processing library that allows access to text, images, and metadata within the document.

The system processes the CV page by page and extracts the following information:

- full textual content of the CV,
- total number of pages,
- embedded hyperlinks such as GitHub and LinkedIn URLs,
- and embedded images, which are used to detect whether a profile photo is present.

The text from each page is combined to create a complete document-level text representation. This allows the system to analyze the CV as a whole instead of processing each page separately.

In addition to text extraction, the system also detects embedded links using PDF link extraction methods. These links are important for validating external references such as GitHub repositories and LinkedIn profiles.

Image detection is performed by scanning each page for embedded images. This helps determine whether the CV contains a profile photo, which is one of the validation criteria.

All extracted elements are stored inside a structured data object called a bundle. The bundle contains:

- extracted full text,

- lowercase text version,
- page count,
- detected links,
- image presence information.

Storing this information in a bundle prevents repeated scanning of the PDF file during validation. This improves processing speed and reduces computational overhead.

3.3.2 Text Normalization

After extracting the text, normalization is applied to prepare the content for accurate validation. Text normalization is necessary because users may write the same information in different formats. For example, "JavaScript", "javascript", and "JAVASCRIPT" all represent the same skill but appear differently in raw text.

To ensure consistent matching, the system applies the following normalization steps:

- converting all text to lowercase to enable case-insensitive matching,
- removing unnecessary whitespace and extra line breaks,
- combining multi-page content into a single unified text representation,
- and cleaning irregular formatting caused by PDF layout variations.

These normalization steps ensure that validation functions can detect keywords, skills, and sections reliably regardless of formatting differences.

This improves the accuracy and consistency of the evaluation process.

3.3.3 Link Normalization and Verification

Links extracted from the CV may appear in different formats, such as:

- github.com/username
- https://github.com/username
- github.com/username/repository

To ensure accurate validation, the system performs link normalization by standardizing link format. This includes:

- adding missing protocols such as https://,
- removing trailing slashes,
- removing unnecessary query parameters,
- and cleaning malformed URLs.

After normalization, GitHub links are verified using HTTP requests. The system sends a request to the GitHub server and checks the response status. If the server returns a successful response (HTTP status 200), the link is considered valid and accessible. If the link returns a 404 or connection error, it is considered invalid. This approach ensures that the system validates real working GitHub profiles and repositories, rather than simply detecting the word "GitHub" in the CV. This aligns with recruiter expectations, as recruiters often check GitHub profiles to evaluate candidate projects and technical skills.

3.4 CV Enhancement system Development and Implementation

The proposed CV validation and enhancement system was developed as a web based application using the Python programming language and the Flask web framework. Flask was selected because it is lightweight, flexible, and well-suited for building scalable web applications with modular architecture.

The system follows a modular design approach, where each validation rule is implemented as an independent function. This modular structure improves maintainability, scalability, and reliability. It also allows new validation rules to be added in the future without affecting existing system components. The system processes CV files through several stages, including secure upload handling, preprocessing, rule-based validation, AI-assisted validation, scoring, and feedback generation.

3.4.1 Platform Architecture

The system architecture consists of several core components that work together to perform CV validation and enhancement.

1. Upload Validation and Security Checks

This component handles user file uploads securely. It verifies file format, validates file size, and ensures that only valid PDF files are accepted.

2. PDF Preprocessing and Bundle Extraction

Once the CV is uploaded, the system extracts text, links, images, and page count using the PyMuPDF library. These extracted elements are stored in a structured data object called a bundle. This bundle is reused by validation functions to improve efficiency and avoid repeated file reading.

3. Rule-Based Validation Module

This component contains multiple validation functions that analyze different aspects of the CV, such as technical skills, formatting, GitHub links, and contact information. Each validation function evaluates one specific CV quality criterion and returns structured feedback.

4. AI-Assisted Validation Layer (LLM)

In addition to rule-based validation, the system uses a Large Language Model (LLM) to perform AI-assisted CV evaluation. The AI analyzes CV content using a structured prompt and returns structured validation results. This provides human-like evaluation and improves validation accuracy.

5. Weighted Scoring and Grading Module

This component calculates the overall CV score by combining results from rule-based validation and AI-assisted validation. Each validation category is assigned a weight based on recruiter importance. The final score is normalized and converted into a grade such as Excellent, Good, Fair, or Needs Improvement.

6. Dimension-Based Scoring Module

This module calculates scores across different quality dimensions, including: Skills quality Impact and achievements, CV brevity, and Formatting and style. This provides more detailed feedback beyond a single overall score.

7. Feedback Generation and Evidence Extraction

The system generates personalized feedback messages based on validation results. Evidence snippets are extracted from the CV to help users understand detected issues. This improves transparency and helps users identify areas for improvement.

8. Data Storage

The system uses SQLite database storage to record evaluation results, system usage statistics, and performance metrics. This allows monitoring system accuracy and user trends.

9. User Interface and API Response Module

The Flask application serves both the web interface and API responses. It allows users to upload CV files, view results, and receive feedback through a browser interface.

3.4.2 Upload Security and File Integrity Controls

To ensure system security and prevent malicious uploads, several security controls were implemented. First, file format validation is performed using binary signature checking. The system checks whether the uploaded file begins with the standard PDF file header %PDF-. This ensures that the file is a genuine PDF and prevents malicious files from being uploaded using fake file extensions. Second, the system sanitizes uploaded filenames using secure filename handling methods. This prevents file path manipulation and injection attacks. Third, the maximum file upload size is limited to 16 megabytes. This protects the system from excessive resource usage and prevents denial-of-service attacks caused by large file uploads. Fourth, uploaded CV files are stored only temporarily during analysis and are automatically deleted after processing. This ensures user privacy and prevents unnecessary data storage. These security measures ensure safe and reliable system operation.

To improve user experience, the system generates temporary preview files that allow users to visually verify the uploaded CV. Each preview file is managed using the following process:

- A unique identifier is generated using a Universally Unique Identifier (UUID).
- The preview file is stored in a dedicated previews directory.
- The preview file is accessible for a limited time.
- A cleanup process automatically removes preview files after 30 minutes.

This cleanup mechanism prevents storage accumulation and ensures that sensitive CV data is not retained longer than necessary. This approach improves both usability and privacy protection.

3.4.3 Rule-Based Validation Modules

The core validation logic of the system is implemented using multiple rule-based validation functions. Each validation function evaluates one specific aspect of the CV.

Each validation function returns structured output containing:

- status – indicates whether the validation passed, failed, or needs improvement
- score – numerical score between 0 and 10
- value – summary result
- message – user-friendly feedback
- optional evidence snippets – extracted text showing detected issues

This structured output allows consistent scoring and feedback generation. The system includes the following validation modules:

CV Length Validation

Implemented using check_cv_page_count(), This function checks whether the CV length is within the recommended range of one to two pages.

GPA Detection

Implemented using check_gpa_in_cv(), This function detects whether GPA information is present and clearly formatted.

Email Validation

Implemented using check_professional_email(), This function verifies the presence and professionalism of the email address.

Profile Photo Detection

Implemented using check_photo_presence(), This function detects whether a profile photo is included in the CV.

School Examination Detection

Implemented using check_ol_al_presence(), This function detects O/L and A/L results and provides guidance based on recruiter preferences.

Formatting Quality Validation

Implemented using check_formatting_quality(), This function evaluates font consistency, margins, and formatting quality.

Technical Keyword Validation

Implemented using validate_technical_keywords(), This function detects relevant technical skills using predefined keyword patterns.

GitHub Link Validation

Implemented using validate_github_links(), This function extracts GitHub links and verifies whether they are accessible and valid.

Skills Section Validation

Implemented using check_skills_separation(), This function detects technical skills and soft skills sections.

Contact Information Validation

Implemented using check_contact_information(), This function verifies phone number, email, LinkedIn, and location information.

Action Verb Validation

Implemented using check_action_verbs(), This function detects strong action verbs that indicate impact and experience.

Quantifiable Achievement Validation

Implemented using check_quantifiable_achievements(), This function detects measurable achievements such as percentages and numeric impact.

Professional Summary Detection

Implemented using check_professional_summary(), This function detects whether a professional summary is present.

3.4.4 AI-Assisted Validation (LLM Validation Layer)

Although rule-based validation ensures consistency and structured evaluation, it has certain limitations. Rule-based systems depend on predefined patterns and keyword matching. If a CV uses unusual formatting, uncommon wording, or creative layout styles, some valid information may not be detected correctly. To reduce this limitation, the proposed system includes an AI-assisted validation layer using a Large Language Model (LLM). This layer provides human-like evaluation by analyzing the CV content more flexibly and interpreting context rather than relying only on strict pattern matching.

The AI-assisted validation does not replace rule-based checks. Instead, it works as a secondary verification layer to confirm whether important CV elements are clearly visible and understandable from a human perspective.

Prompt Preparation Strategy

To ensure reliable and consistent AI responses, the system uses a fixed and structured prompt. The same prompt format is used for every CV evaluation to maintain fairness and repeatability.

The prompt includes the following requirements:

- The AI must return STRICT JSON output only, without any explanations or additional text.
- The output must follow a predefined structure.
- The AI must evaluate exactly nine predefined criteria.

By limiting the evaluation to predefined criteria and requiring structured JSON output, the system ensures that AI responses are easy to parse programmatically and integrate into the scoring system.

AI Score Calculation

After the AI returns structured JSON output, the system parses the response safely and extracts the results for the nine evaluation criteria. Each criterion contains a Boolean value (yes: true or yes: false). The system counts how many criteria passed (i.e., how many "yes" values are true).

The AI score is calculated using the following formula:

LLM Score = (Passed Criteria / 9) × 10

For example, if 7 out of 9 criteria are satisfied:

LLM Score = (7 / 9) × 10 = 7.8

The final LLM score is rounded to one decimal place.

Fairness and Bias Reduction (Blind Mode)

To support ethical evaluation and reduce bias, the system includes a feature called Blind Mode.

When Blind Mode is enabled:

- Profile photo is ignored in scoring.
- O/L and A/L results are treated as low importance.
- Photo score is neutralized to prevent visual bias.

This is important because appearance, school background, or personal attributes should not negatively influence technical hiring decisions.

Blind Mode ensures:

- Fair evaluation
- Reduced unconscious bias
- Better suitability for screening environments

This feature supports responsible AI practices and fairness in recruitment technology.

3.4.5 Feedback Generation Mechanism

The system does not only generate scores; it also provides structured feedback.

Each validation function returns:

- Status (success, warning, info, error)
- Score (0–10)
- Summary value
- Clear message
- Evidence snippets (when applicable)

Feedback is designed to be:

1. Clear – simple English, easy for undergraduates
2. Actionable – tells exactly what to fix
3. Evidence-based – shows where the issue exists

For example:

- "Only 2 action verbs found. Use words like 'Developed', 'Implemented'."
- "GitHub link not working"

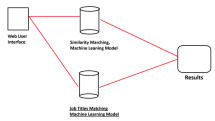
Evidence snippets are extracted using pattern matching so that users can locate the problem in their CV quickly. Finally, the system generates a summary feedback message based on the overall score and grade, helping users understand their general performance level.

3.5 Job Recommendation System Development and Implementation

This chapter explains the implementation of the Job Recommendation part, which is part of the CV Enhancement and Job Opportunity Recommendation System. This part uses a multi-model architecture to provide more accurate job matching for IT job seekers. Currently, the system combines two machine learning approaches: one Similarity Matching Machine Learning Model for job searching via keyword matching with the job post, and another Job Titles Matching Machine Learning Model for predicting job titles. Both models run in parallel to generate more reliable job recommendations.

3.5.1 System Architecture

The Job Recommendation Module is built around a multi-model method. When a user submits their CV through the Web User Interface, the system sends the input to both models at the same time. The first model checks whether the skills listed in the user's CV match job postings collected from TopJobs.lk. The second model analyzes the same input and predicts the most suitable IT job titles based on the user's skills. It then checks whether jobs related to those predicted titles exist in the job postings from TopJobs.lk. The outputs from both models are combined and displayed to the user as a list of job recommendations. This parallel execution helps improve the accuracy of the recommended jobs.



3.5.2 Dataset Preparation

During the job dataset preparation phase, job advertisements were collected from Sri Lankan job portals such as TopJobs.lk using web scraping techniques. However, during initial data collection it was observed that a significant number of job advertisements were posted as image-based flyers instead of structured HTML text. As a result, traditional scraping methods were unable to extract important information such as required skills, job descriptions, and qualifications.

This limitation caused incomplete datasets and reduced the accuracy of the job recommendation engine because the system could not detect key skill requirements from these image-based advertisements. To overcome this issue, an Optical Character Recognition (OCR) based extraction pipeline was introduced.

First, the system identifies job advertisements that contain embedded images or flyer-style posts. These images are then processed using EasyOCR, which converts the visual text within the image into machine-readable textual content.

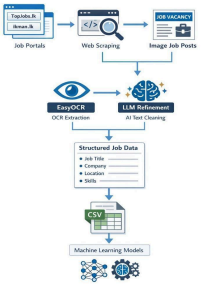
However, OCR-generated text often contains noise, formatting errors, and incomplete sentences due to variations in fonts, backgrounds, and image quality. To improve the quality of extracted text, a Large Language Model (LLM) refinement stage was introduced.

The LLM processes the raw OCR output and performs the following tasks:

- Correct spelling and formatting errors
- Remove irrelevant text fragments
- Reconstruct broken sentences
- Extract structured information such as job title, required skills

After refinement, the cleaned required skills are stored in the dataset and used as input for the job similarity matching and recommendation models.

This hybrid approach significantly improved the completeness and quality of the job dataset, enabling the recommendation engine to detect required technical skills more accurately and generate more relevant job recommendations.



- **Dataset Preparation for "Similarity Matching, Machine Learning Model."**

The Similarity Matching Machine Learning Model does not require a separate dataset. This is because the model uses the Cosine Similarity Algorithm to check whether the skills listed in the user's CV match the job postings collected from TopJobs.lk. Therefore, no additional dataset is needed for this model.

- **Dataset Preparation for "Job Titles Matching Machine Learning Model."**

The job recommendation model was trained using a structured dataset named IT_Job_Roles_Skills.csv. The dataset contains 493 IT job records and four attributes, as shown below.

Table 1: Data set structure Job Titles Matching Machine Learning Model

Column	Description
Job Title	The name of the IT job role
Job Description	A text description of the role and responsibilities
Skills	A comma-separated list of required technical and soft skills
Certifications	Relevant certifications associated with the role

The dataset covers a wide range of IT roles, including software development, data science, DevOps, AI and machine learning engineering, system administration, cybersecurity, project management, and creative technology roles. Sample job titles from the dataset include Admin Big Data, Ansible Operations Engineer, Artifactory Administrator, Artificial Intelligence / Machine Learning Leader, AI Researcher, AI Software Architect, and Business Intelligence Developer, among 493 total records.

The Skills column was selected as the primary input feature for the Job Titles Matching Model. And the job title was used as the target output.

3.5.3 Data Preprocessing

3.5.3.1 Model 1: "Similarity Marching, Machine Learning Model."

No preprocessing is required.

3.5.3.2 Model 2: "Similarity Marching, Machine Learning Model."

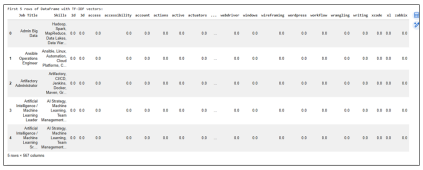
Feature Extraction Using TF-IDF

The Skills column contains skill lists written in different formats across records. To convert this text into a numerical format suitable for machine learning, TF-IDF (Term Frequency-Inverse Document Frequency) vectorization was applied.

The TF-IDF vectorizer was configured with the following settings:

- lowercase=True: All text is converted to lowercase for consistency.
- stop_words='english': Common English stop words are removed to reduce noise.

After fitting the vectorizer on the Skills column, a feature matrix of 565 vocabulary terms was produced across 493 records. Each record is represented as a vector of 565 TF-IDF weighted term scores.



```
First 5 rows of DataFrame with TF-IDF vectors:
                                   Job Title  ...  accessibility  account  actions  active  activity  ...  web3/nft  windows  wireframing  wordpress  workflow  wrangling  writing  xcode  zi  zebbie
0                                Admin Big Data  ...            0.0      0.0      0.0     0.0       0.0  ...       0.0      0.0          0.0        0.0       0.0        0.0      0.0    0.0  0.0     0.0
1                          Ansible Operations Engineer  ...            0.0      0.0      0.0     0.0       0.0  ...       0.0      0.0          0.0        0.0       0.0        0.0      0.0    0.0  0.0     0.0
2                             Artifactory Administrator  ...            0.0      0.0      0.0     0.0       0.0  ...       0.0      0.0          0.0        0.0       0.0        0.0      0.0    0.0  0.0     0.0
3                   Artificial Intelligence / Machine Learning Leader  ...            0.0      0.0      0.0     0.0       0.0  ...       0.0      0.0          0.0        0.0       0.0        0.0      0.0    0.0  0.0     0.0
4                   Artificial Intelligence / Machine Learning Sr. ...  ...            0.0      0.0      0.0     0.0       0.0  ...       0.0      0.0          0.0        0.0       0.0        0.0      0.0    0.0  0.0     0.0

[5 rows x 565 columns]
```

Figure 8:Feature Extraction Using TF-IDF

Label Encoding

The target variable Job Title contains 493 distinct string class labels. Label encoding was applied using scikit-learn's LabelEncoder to convert these string labels into integer class indices. This is required for training classification models that expect numeric target values.

Train-Test Split

The dataset was split into a training set and a test set using a 70/30 ratio:

Table 2:Train-Test Split

Split	Records
Training Set (70%)	345 samples
Test Set (30%)	148 samples

```
x_train.shape
(345, 565)
x_test.shape
(148, 565)
```

Figure 9:Train-Test Split

3.5.3.3 Model 3: Similarity Matching Machine Learning Model

The Similarity Matching model uses Term Frequency-Inverse Document Frequency (TF-IDF) vectorization combined with Cosine Similarity scoring. TF-IDF assigns numerical weights to words based on how frequently they appear in a given document relative to how commonly they appear across the entire corpus. Cosine Similarity then measures the angle between two TF-IDF vectors in high-dimensional space, producing a score between 0 and 1 where values approaching 1 indicate strong textual alignment between the user's skills and a job posting.

Vectorization Configuration

The TF-IDF vectorizer was configured with a maximum of 500 features and English stop word removal. The user's skill input and all job listing texts were jointly vectorized into a single matrix, ensuring that term weights are computed within the same document space for a fair comparison.

Similarity Computation and Ranking

The user's skill vector, positioned as the last row of the TF-IDF matrix, is compared against all job listing vectors using cosine similarity. The resulting similarity scores are appended to the job listings data frame, which is then sorted in descending order. The top-N highest-scoring listings are returned as recommendations.

Location-Based Filtering

Before similarity computation is performed, an optional location filter is applied. When the user selects a specific location from the interface, the job listing data frame is pre-filtered using a case-insensitive string match on the location field. This ensures that the similarity scoring operates only on geographically relevant postings, improving the practical usefulness of the recommendations.

3.5.3.4 Model 4: Job Titles Matching Machine Learning Model

The Job Titles Matching model approaches recommendation as a multi-class classification problem. Rather than comparing text strings directly, this model has been trained to recognize which job title category best matches a given combination of skills. Six different classification algorithms were implemented and evaluated in parallel, and their predictions are presented together to give the user a broader view of suitable job title categories.

Algorithms Used

Table 3:Algorithms used

Model	Configuration
Naive Bayes	MultinomialNB (default settings)
Support Vector Machine	Linear kernel, probability=True
Random Forest	100 estimators, random_state=42
Logistic Regression	max_iter=1000
k-Nearest Neighbors	k=3
Decision Tree	random_state=42

All six models were trained on the full TF-IDF feature matrix using the label-encoded Job Title as the target class.

In addition to the six ML models, a TF-IDF Cosine Similarity baseline was implemented. This method directly computes the cosine similarity between the candidate's input skill vector and every record in the training dataset, returning the top matching job titles by similarity score.

3.5.4 Web Interface

To see the recommending jobs, user need to click on "Job Recommendation" button. After that he will redirects to this page.

Upon processing the user's CV, the interface displays a structured results page that combines the outputs of both the Similarity Matching model and the Job Titles Matching model into a single, user-friendly view.

Job recommendations are then filtered and tagged according to seniority level, with the interface providing four selectable tabs: All Levels, Intern / Junior, Mid-Level, and Senior / Lead. In the example, 13 total jobs were identified 9 at Intern / Junior level, 3 at Mid-Level, and 1 at Senior / Lead allowing the user to focus on opportunities appropriate to their experience.

Each recommended job is presented as an individual card containing the job title with a color-coded career level badge, the employer's name and the location, the application closing date, and a match percentage score indicating the degree of alignment between the user's CV and the job posting. Match scores are classified into three tiers: Excellent for scores of 70% and above, Good for scores between 50% and 69%, and Potential for scores below 50%. A distinguishing feature of the interface is the integrated skill gap analysis presented within each job card. Rather than simply listing matching positions, the system explicitly identifies both the technical and soft skills that the user currently possesses and those that are required but absent from their CV. In the example shown in Figure, the user matched seven technical skills against the SE Intern Development position at Infor, while five technical skills were flagged as missing, alongside a soft skill match ratio of 2 out of 5. This functionality transforms the job recommendation output from a passive list into an active career development tool, enabling users to understand precisely what skills they need to acquire to strengthen their candidacy for each role. The interface additionally provides a "Validate Another CV" button to restart the process, an "Apply Now" button on each card linking directly to the original job posting, and a back navigation control, ensuring a smooth and self-contained user experience throughout.

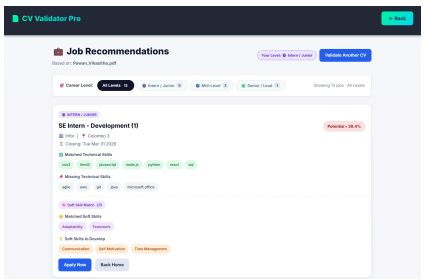
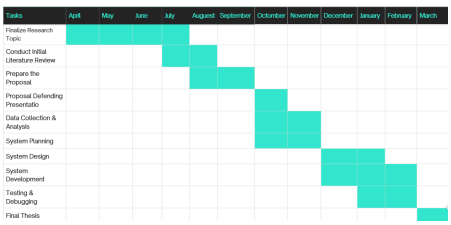


Figure 10 : web interface

Project Timeline



4. Results and Evaluation

This chapter presents the experimental results and performance evaluation of the proposed AI-assisted CV enhancement and job opportunity recommendation system. The CV evaluation was conducted to assess both the technical accuracy of the system and its practical usefulness for real-world users.

The system consists of two main components:

1. CV Enhancement Module
2. Job Recommendation Module

When a user uploads a CV, the system extracts relevant information such as skills, education, contact details, etc. These extracted features are then used for both CV evaluation and job recommendation. The system applies multiple machine learning algorithms and similarity-based techniques to analyze the input and generate results.

Since the system uses a multi-model approach, the prediction results may vary depending on the input CV content. Therefore, a real-world testing scenario was used to evaluate system performance and ensure reliability. In the job recommendation process, when a user uploads a CV, the system extracts the skills from the CV and checks them using several classification algorithms. The algorithms try to match the skills with job categories. The algorithm that produces the highest accuracy is selected for the final prediction.

Because of this approach, the accuracy can change depending on the input data. Therefore, to demonstrate the system performance, we used a real scenario and ran a test. The results obtained from that test are explained below.

4.1 Model Accuracy Results

All six classifiers were trained on the grouped dataset and evaluated using training accuracy, test accuracy, and 5-fold stratified cross-validation.

Table 4:Grouped dataset

Job Category	Samples
Software Development	92
DevOps & Cloud	52
Data Science & AI	43
IT Management & Support	27
Cybersecurity	19
Network Engineering	15
Project & Product Management	10
IT Architecture	9
QA & Testing	8
Database Administration	7
Business Analysis	7
Technical Leadership	3


```
Original dataset: 493 records, 469 unique job titles

Job Category Distribution:
------------------------------------------------------------
  Other IT Roles                      200 samples
  Software Development                 92 samples
  DevOps & Cloud                       52 samples
  Data Science & AI                    43 samples
  IT Management & Support              27 samples
  Cybersecurity                        19 samples
  Network Engineering                  15 samples
  Project & Product Management         10 samples
  IT Architecture                       9 samples
  QA & Testing                          8 samples
  Database Administration               7 samples
  Business Analysis                     7 samples
  Technical Leadership                  3 samples
  Frontend Development                  1 samples

Total categories: 14
Removed 'Other IT Roles': 200 samples

Clean dataset: 292 records, 12 categories


  Clean dataset: 292 records, 12 categories

  Training samples : 204
  Testing samples  : 88

  ================================================================================
    Model                    Train Acc   Test Acc    CV (5-fold)
  ================================================================================
    Naive Bayes               76.5%      72.7%      73.6% ±3.5%
    SVM                       94.1%      79.5%      82.9% ±3.3%
    Random Forest            100.0%      75.0%      76.1% ±4.0%
    Logistic Regression       86.3%      75.0%      79.4% ±4.8%
    KNN                       85.7%      77.3%      80.5% ±2.5%
    Decision Tree            100.0%      63.6%      76.4% ±6.3%
  ================================================================================

  Best Model (Test Accuracy): SVM — 79.5%


  Best Model (Test Accuracy): SVM — 79.5%

  ================================================================
    CLASSIFICATION REPORT — SVM
  ================================================================
                                  precision    recall  f1-score   support

                   Business Analysis       0.50      0.50      0.50         2
                       Cybersecurity       0.71      0.83      0.77         6
                  Data Science & AI        0.81      1.00      0.90        13
          Database Administration          0.00      0.00      0.00         2
                     DevOps & Cloud        0.78      0.88      0.82        16
                  Software Development     0.86      0.75      0.80        24
          IT Management & Support          0.64      0.88      0.74         8
                  Network Engineering      0.75      0.75      0.75         4
      Project & Product Management         1.00      0.67      0.80         3
                       QA & Testing        1.00      0.50      0.67         2
                  Software Engineer        0.80      0.80      0.80        15
                 Technical Leadership      0.00      0.00      0.00         1

                             accuracy                           0.80        89
                            macro avg      0.67      0.60      0.61        89
                         weighted avg      0.79      0.80      0.78        89
```

Figure 11:Model Accuracy Results

The Support Vector Machine achieved the highest test accuracy of 79.5% and the highest cross validation score of 82.9% ±3.3%, making it the best-performing classifier for this task. The relatively low standard deviation in the CV score indicates consistent performance across different data splits, which is a reliable indicator of generalizability. The linear kernel configuration of the SVM is well-suited to TF-IDF feature spaces, where classes are often linearly separable in high-dimensional sparse representations.

KNN achieved the second highest test accuracy at 77.3% with a CV score of 80.5% ±2.5% the lowest variance across all models, indicating stable predictions. Random Forest followed with 76.1% test accuracy, though its perfect training accuracy of 100% reveals a degree of overfitting, as the ensemble memorized training samples without fully generalizing to unseen inputs.

Logistic Regression and Naive Bayes recorded test accuracies of 73.9% and 72.7% respectively, both performing reasonably well given the class imbalance present in the dataset. The Decision Tree classifier recorded the lowest test accuracy at 63.6% despite achieving 100% training accuracy the largest train-test gap among all models confirming its susceptibility to overfitting on sparse, high-dimensional text features. This behavior is consistent with findings reported in the literature on single decision tree classifiers applied to text classification problems.

The overall accuracy range of 63.6% to 79.5% across all six models demonstrates that the multi-algorithm approach is well-justified. Rather than committing to a single classifier, the system evaluates all six models at inference time and selects the prediction with the highest confidence score, effectively leveraging the complementary strengths of different algorithms.

4.2 Final System Evaluation (Web Integration)

4.2.1 Web Application Testing

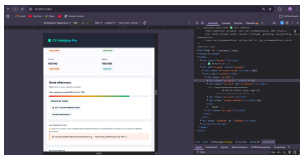
The web application developed for the CV validation and job recommendation system was tested across multiple web browsers and devices to ensure compatibility, usability, and reliability. The application was tested on popular browsers including Google Chrome, Mozilla Firefox, and Microsoft Edge. Testing was conducted on different devices such as Windows-based desktop computers, laptops, and tablets to confirm that the system functions correctly in various environments.

The web interface allows users to upload their CV files easily through a simple file upload option. After uploading the document, the system processes the CV through the validation pipeline and generates feedback along with job recommendations. The results are displayed on the results page, where users can clearly view the CV quality score, extracted skills, improvement suggestions, and recommended job opportunities.

Key Features Tested:

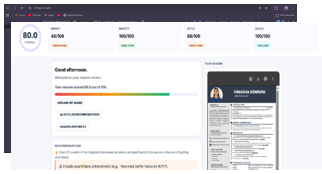
• Responsiveness

The web application was designed using responsive layout techniques and custom CSS styling. During testing, the interface adapted effectively to different screen sizes, ensuring that the application remained user-friendly on both large desktop monitors and smaller devices such as laptops and tablets.

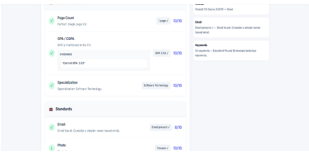


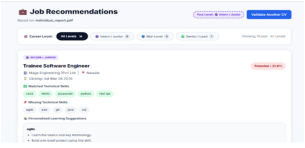
• CV Analysis and Job Recommendation Results:

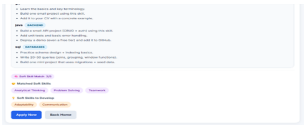
After uploading the CV, the system processes the document and displays detailed validation feedback. This includes information such as CV structure evaluation, detected skills, and suggestions for improving the CV. Based on the extracted skills, the system generates job



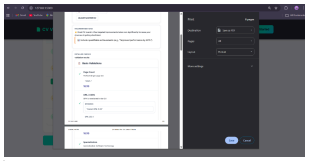
recommendations using job listings collected from platforms such as TopJobs.lk.







• System Performance:



The response time of the system was measured during testing to evaluate performance. On average, the CV analysis and job recommendation process was completed within a few seconds

depending on the size and complexity of the uploaded document. This performance level enables near real-time feedback for users, ensuring a smooth user experience.

4.2.2 Evaluation Metrics and Results

To quantitatively evaluate the performance of the developed system, several CV documents were tested through the complete workflow. These CVs were uploaded to the web application and processed through the CV validation pipeline and job recommendation engine. The evaluation focused on measuring system accuracy, response time, reliability, and overall user experience.

During testing, the system successfully extracted relevant information such as technical skills, education details, and contact information from most CV documents. The job recommendation engine was then able to generate suitable job opportunities based on the extracted skills by comparing them with job listings obtained from platforms such as TopJobs.lk. The evaluation confirmed that the system can effectively analyze CV content and provide relevant job recommendations.

Table 5:Evaluation Metrics and Results

Metric	Value
CV information extraction accuracy	75-85%
Relevant job recommendation accuracy	70-80%
Average system response time	5-15 Seconds
User satisfaction (UI feedback)	Good